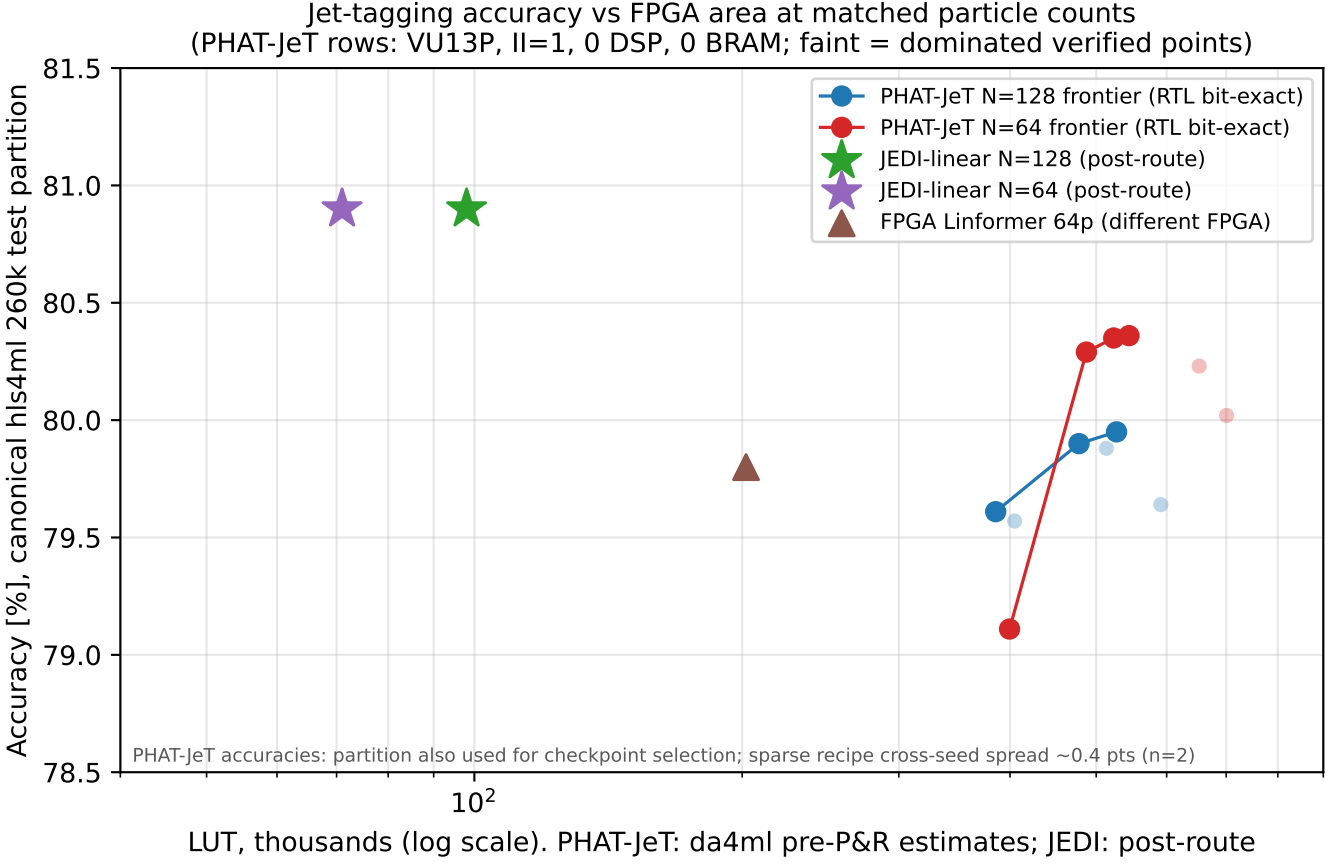
Jet-tagging accuracy vs FPGA area at matched particle counts
(PHAT-JeT rows: VU13P, II=1, 0 DSP, 0 BRAM; faint = dominated verified points)
PHAT-JeT N=128 frontier (RTL bit-exact)
PHAT-JeT N=64 frontier (RTL bit-exact)
JEDI-linear N=128 (post-route)
JEDI-linear N=64 (post-route)
FPGA Linformer 64p (different FPGA)
Accuracy [%], canonical hls4ml 260k test partition
81.5
81.0
80.5
80.0
79.5
79.0
78.5
10^2
PHAT-JeT accuracies: partition also used for checkpoint selection; sparse recipe cross-seed spread ~0.4 pts (n=2)
LUT, thousands (log scale). PHAT-JeT: da4ml pre-P&R estimates; JEDI: post-route